Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Smith, Fredrick D B.S.		
Project Title:	Sabatier Risk Mitigation and Development		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	10/01/2001	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	Carbon dioxide reduction is not baselined for inclusion in the Air Revitalization of the International Space Station. However, the Oxygen Generation System (OGS) rack has been scarred for later installation of a CRA. A team consisting of the NASA Marshall Space Flight Center (MSFC), Johnson Space Center (JSC), AMES Research Center (ARC), Southwest Research Institute (SWRI) and Hamilton Sundstrand have been working together to understand requirements for a CRA in order to define the scar in the OGS rack. Risk mitigation activities have been performed to elevate the CRA Technology Readiness Level (TRL) to TRL 6. One of the areas of risk is the integration of the CRA with the CDRA and the OGA. The Sabatier operates on the same 90-minute diurnal cycle as the OGA, which is different than the 144-minute cycle of the CDRA. One of the objectives of the development of a Sabatier EDU is to use it for integration testing with the development OGA and the development CDRA. A development CO2 compressor is being developed in parallel by SWRI that will be tested with the CDRA and the Sabatier EDU. The Sabatier EDU has been designed to be		

Task Description:	functionally equivalent to the Sabatier Reactor Subsystem (SRS) portion of the CRA. The Sabatier Engineering Development Unit (EDU) (formerly the Systems Integration Research Faciltiy (SIRF) or the Lunar Mars Life Support Project (LMLSTP)CO2 Reduction Assembly) has been refurbished as part of the Sabatier Risk Mitigation and Development activity. The system is designed to be functionally equivalent to a Carbon Dioxide Reduction Assembly (CRA) for the International Space Station (ISS). The CRA consists of a Sabatier Reactor Subsystem (SRS) and a Carbon Dioxide Management Subsystem (CMS). The EDU reflects only the SRS portion of the CRA; the carbon dioxide compressor is being developed separately. The SRS includes a Sabatier methanation reactor, condensing heat exchanger, phase separator and the associated valves and sensors required for safe operation. The purpose of the EDU is to test the control algorithms being developed for the CRA and to understand the interaction between the CRA and the Oxygen Generation Assembly (OGA) and the Carbon Dioxide Removal Assembly (CRDA).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Sabatier development presence potential benefits with CO2 recovery in industrial applications which would minimize the CO2 released to the environment and recover water that has many impacts. While providing Hydrogen in an effecient manner is key to this technology for the recovery of the water, with advancements in inexpensive/efficient H2 recovery the Sabatier technology can benefit greatly industry and other earth based activities (environmental systems, remote operations,etc.)
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)